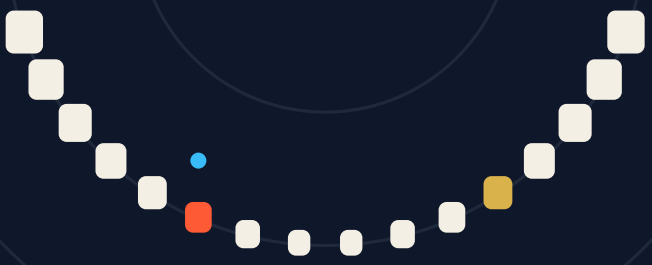

HACKATHON IDEA BOOK · DENTAL 3D

ToothTwin

A living 3D twin of the mouth that you can talk to, learn on and treat in.

LEARN

AI virtual patients for dental students

SEE

Your diagnosis, explained on your own 3D mouth

TEACH

Live classes and AI-written cases for lecturers

Project review, feature ideas by user type, model roadmap, demo script and build plan

November 2026

01 · PROJECT REVIEW

What you already have

Most hackathon teams spend the weekend just getting a 3D model on screen. You start with a complete, original, interactive dental mouth. That is your unfair advantage: spend the hackathon on the **intelligence and the story**, not on the 3D plumbing.

Asset in the repo	Why it matters for judges
Procedural 3D mouth (32 teeth, gums, bone, tongue)	Generated from signed distance fields in *scripts/dental-model*, so there is **no licensed model**. It is fully parametric: you can regenerate teeth with different shapes, ages or disease.
5 view layers: gums, bone, roots, teeth, X-ray	Already lets you "see through" the patient. A perfect base for teaching and for patient explanations.
Tooth cross-section (enamel, dentin, pulp)	Stencil-capped solid section is a genuinely impressive visual most people have never seen interactively.
Surface picking + FDI odontogram + primary teeth	Real clinical data model: conditions, surfaces, Palmer aliases, quadrants and soft-tissue sites.
Site insight data (structures, differentials, tests)	A ready-made knowledge base for an AI tutor or diagnostic copilot to ground its answers in.
Case form that already builds a JSON payload	The hook for an AI work-up is already there: today it just prints the payload on the page.

Gaps to close for a winning demo

- **No AI yet.** The case payload stops at the screen. Connecting it to an LLM is the single biggest upgrade.
- **No story or user.** It is a powerful tool without a hero. Judges remember a person with a problem.
- **Static disease.** Conditions are shown as colour tints. Showing them *physically* (a carved cavity, receding gum) is far more striking.
- **Input is mouse-only.** Talking to the AI, or reading an X-ray, would make the demo feel like magic.

The big idea: turn the 3D mouth into a dental twin that an AI can talk about, mark up and animate, then put it in the hands of dental students, their teachers and the patients they will one day treat.

02 · FEATURES BY USER

Who it is for, section by section

Pick one persona as the hero of your pitch (we recommend the **student**, because the learning angle is safe, measurable and loved by judges) and show the others as "and it also works for...".

Dental student

Learning anatomy, diagnosis and exams. Hero persona: biggest pain, easiest to demo, no clinical risk.

AI virtual patient (case simulator)

Medium · WOW ●●●

An AI plays a patient with a hidden diagnosis: "My lower left tooth hurts when I drink cold water." The student chats to take a history, picks teeth on the 3D mouth, orders tests (cold test, percussion, radiograph) and gets realistic results back. They commit to a diagnosis and get scored with feedback.

How: LLM with a hidden case card. Reuse DENTAL_INSIGHTS for valid tests and differentials. Tests return scripted findings.

"Find the tooth" anatomy quiz

Quick · WOW ●●○

"Click tooth 36." "Show me the mesial surface of UR4." "Where is the inferior alveolar nerve?" Timed rounds with streaks and a class leaderboard. Switches between FDI, Palmer and Universal notation.

How: Your picking and tooth catalogue already exist. Add a question generator and a score state.

Ask-the-tooth AI tutor

Medium · WOW ●●●

Select any tooth or layer and ask "Why does this one have two roots?" The tutor answers and the camera flies to and highlights the structures it talks about.

How: LLM returns text plus a list of tooth ids and layer names; CameraRig animates to them.

OSCE exam mode

Quick · WOW ●●○

A timed mock exam station: chart a full mouth from a case description, then auto-mark the odontogram against the answer key.

How: Compare the student's chart JSON to the key, tooth by tooth and surface by surface.

Weak-spot tracker

Quick · WOW ●○○

Tracks every wrong answer (e.g. always confuses premolars) and builds the next quiz from those mistakes, like spaced repetition.

Talk to the patient by voice

Quick · WOW ●●●

Students take the history out loud, as in a real clinic, and the virtual patient answers out loud. Practises communication, not just knowledge, and looks magical on stage.

How: Browser Web Speech API for speech to text, and speechSynthesis for the patient's voice. No extra service.

X-ray reading trainer

Stretch · WOW ●●●

Show a radiograph; the student marks what they see (caries, bone loss, a root filling) on the matching 3D teeth. The AI checks the answer and explains what they missed.

How: Start with a multimodal LLM and a few labelled images; later use a public panoramic dataset (e.g. DENTEX).

Patient

Scared, confused, and told "you need a root canal" with no picture of what that means.

"Explain my treatment" in 3D

Medium

WOW ●●●

Pick a diagnosis (or type what your dentist told you) and it becomes a short guided animation on the 3D mouth: the cavity, the drill path, the filling or root canal, step by step, narrated in plain language (and in the patient's own language).

How: Use the existing section view and clipping planes; LLM writes the narration from the diagnosis; browser speech reads it out.

Symptom finder

Quick

WOW ●●○

"It hurts at the back on the left when I chew." The AI highlights the likely teeth on the 3D mouth and gives triage advice (see a dentist this week vs. go to urgent care now), with a clear disclaimer.

Future-you slider

Stretch

WOW ●●●

"What happens if I wait?" Slide time forward and watch the cavity grow into the pulp, or gums recede. A powerful motivation tool for brushing, flossing and booking that check-up.

Teacher / university

Wants to run classes and check who is actually learning.

Live classroom mode

Medium

WOW ●●○

The lecturer rotates the model and every student's screen follows; quick polls ("which tooth is this?") show live results.

Case authoring with AI

Quick

WOW ●●○

Teacher types "a 45-year-old with a cracked tooth"; the AI generates a full virtual patient case for the simulator.

Class analytics dashboard

Medium

WOW ●○○

Which topics the class gets wrong most, average diagnosis accuracy, time per case.

03 · MODEL ROADMAP

Advancing the models

Two models can grow: the **3D model** (your SDF generator) and the **AI model**. Because your mouth is generated from math rather than sculpted, you can do things a downloaded model never could. Say this out loud to the judges: it is a technical moat.

A. The 3D model (signed distance fields)

Upgrade	What it unlocks	Effort
Carved disease	Subtract a small noisy sphere from the enamel/dentin field to carve a real cavity; extrude for calculus; lower the gum field for recession. Disease becomes 3D, not a colour.	Medium
Age morphing	Interpolate between the primary and permanent dentitions you already catalogue: watch baby teeth fall out and adult teeth erupt from age 6 to 21. Superb for students and parents.	Stretch
Orthodontic simulation	Move and rotate teeth along the arch to show crowding, then animate aligner stages back to a straight smile.	Medium
Bite heatmap	Close the jaw (jaw control exists) and colour where upper and lower teeth touch: occlusion analysis.	Medium
Patient-specific shape	Fit the SDF parameters (arch width, tooth sizes) to a photo or a scan so the twin looks like *this* patient.	Stretch
Nerves and vessels layer	Add the inferior alveolar nerve canal and maxillary sinus as a new layer: key for wisdom-tooth and implant planning.	Medium
AR / WebXR view	Place the mouth on a desk through a phone camera. Pure wow for a stage demo.	Stretch

B. The AI model

Stage	Approach	Hackathon fit
1. Prompted LLM	A strong LLM (e.g. Claude) with your DENTAL_INSIGHTS and TOOTH_CONDITIONS in the prompt, returning structured JSON (tooth ids, conditions, camera targets) via tool calling.	Build this weekend
2. Grounded (RAG)	Index dental textbooks or guidelines so every answer cites a source. Judges love "no hallucinated medicine".	If time allows
3. Vision	Multimodal model reads X-rays and photos, maps findings to FDI numbers on the twin.	Stretch demo
4. Specialised	Fine-tune a tooth detection / caries segmentation model on public panoramic X-ray datasets; keep the LLM for reasoning and explanation.	Future roadmap slide
5. Learning loop	Student answers and teacher corrections become labelled data that improves the tutor and detector over time.	Future roadmap slide

04 · THE DEMO THAT WINS

A 3-minute demo script

Judges decide in the first 30 seconds. Lead with a person, show magic early, end with the vision.

Time	On screen	What you say
0:00	Black screen, then the 3D mouth fades in and slowly rotates.	"Dental students learn 3D anatomy from 2D textbooks, and patients say yes to treatments they cannot picture."
0:20	Click "Student". The AI patient sends: "My bottom left tooth hurts when I drink cold water."	"Meet Sara, a virtual patient. Our student has to find out what is wrong."
0:50	Student asks questions, clicks tooth 36, orders a cold test, peels the gums with the layer switch, opens the cross-section.	"Every tooth, root and nerve is explorable. And we generated the whole mouth from math, with no licensed model."
1:30	Student submits "reversible pulpitis". Score card and feedback appear.	"Instant feedback, like a tutor sitting next to you."
1:50	Student asks out loud: "Why does this tooth have two roots?" The camera flies to tooth 36, peels the bone and highlights both roots while the tutor answers.	"Ask anything, and the mouth answers by showing you." (Pause here. Let the room react.)
2:20	Click "Explain to patient". The cavity is carved, filling animates in, narration plays.	"And the same twin explains a filling to a nervous patient, in their own language."
2:40	Roadmap slide: X-ray vision, age morphing, AR.	"Learn it, teach it, understand it. This is ToothTwin."

Safety line for the pitch

"ToothTwin is an education tool. Every case is a synthetic patient, and the patient mode explains treatments but never diagnoses: it always says to see a dentist." Saying this early removes the judges' biggest objection to health AI.

Demo survival tips

- **Record a backup video** of the full demo. Wi-Fi and microphones fail on stage.
- **Pre-warm the AI**: have one case cached so the first response is instant.
- **Use a scripted voice line** you have tested 20 times, in a quiet corner if possible.
- **Keep the mouse moves big and slow**; the projector audience cannot follow small clicks.

05 · BUILD PLAN

What to build, in what order

Block	Goal	Done when
Hours 0-3	Wire the LLM. Replace the payload print in CaseForm with an API call that returns structured JSON.	The AI returns differentials and tooth ids; the mouth highlights them.
Hours 3-9	Student case simulator: chat, hidden case card, test results, scoring screen.	One full case can be played start to finish.
Hours 9-13	AI tutor that drives the scene: answers come back with tooth ids and layers, and the camera flies to them. Add voice input.	Asking a spoken question moves the camera and highlights the right teeth.
Hours 13-18	Patient explain mode: narration plus camera fly-through and the section view.	One treatment (filling) explained end to end.
Hours 18-21	Polish: landing screen with the three personas, transitions, loading states.	A stranger can use it without help.
Hours 21-24	Pitch deck, backup video, rehearse the demo at least 5 times.	Demo fits in 3:00 with no surprises.

Suggested stack additions

- **AI:** Anthropic API (e.g. *claude-sonnet-5-5*) with tool use so answers come back as typed JSON your scene can act on.
- **Backend:** a tiny serverless function to keep the API key off the client.
- **Voice:** the browser Web Speech API for input and speechSynthesis for narration: zero cost, no extra service.
- **Animation:** GSAP or react-spring timelines for camera fly-throughs and the treatment animation.

If you only build three things

1. AI virtual patient 2. AI tutor that moves the 3D mouth 3. 3D treatment explanation for patients. Everything else is the roadmap.

06 · WINNING THE ROOM

How it scores with judges

Typical criterion	How ToothTwin hits it
Innovation	AI that does not just talk but **acts on a 3D twin**: highlights, carves and animates. Plus a procedurally generated anatomy no one else has.
Technical complexity	SDF mesh generation, stencil-capped cross-sections, meshopt compression, LLM tool calling, speech input.
Impact	Dental education worldwide (students, universities, self-learners) and better patient understanding of treatment.
Design / UX	Already polished UI; clear personas; spoken conversation and visual feedback.
Feasibility / business	B2B licences for dental schools, a premium plan for exam prep, a free tier for students.
Presentation	A story with a virtual patient, one jaw-drop moment (the tutor driving the 3D mouth) and a clear vision slide.

Lines to use in the pitch

"Students learn a 3D organ from 2D pictures. We fixed that."

"Our mouth was generated from mathematics, so we can give it any disease, any age, any patient."

"The AI does not just answer. It points, carves and animates."

"One dental twin: the student learns on it, the teacher teaches with it, the patient finally understands."

Risks and how to handle them

Risk	Mitigation
Judges worry about medical AI safety	Lead with education; synthetic cases only; patient mode never diagnoses; disclaimers on screen.
Live AI is slow or wrong on stage	Cache the demo case, stream responses, keep a backup video.
Scope creep	Ship the three "Quick" wow features first; everything "Stretch" goes on the roadmap slide.
Patient data / privacy questions	Synthetic cases only in the demo; mention on-device speech and no stored PHI.